RISE RESEARCH INNOVATION SCHOLARSHIP ENTREPRENEURSHIP

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Subject/Category : Machine Learning
Class : DS 4420

SCAN ME

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[redacted] into a Car: Detecting Driver Distraction with Machine Learning

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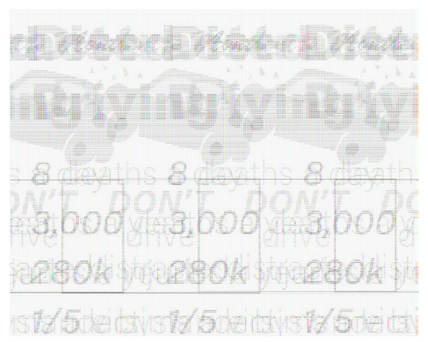
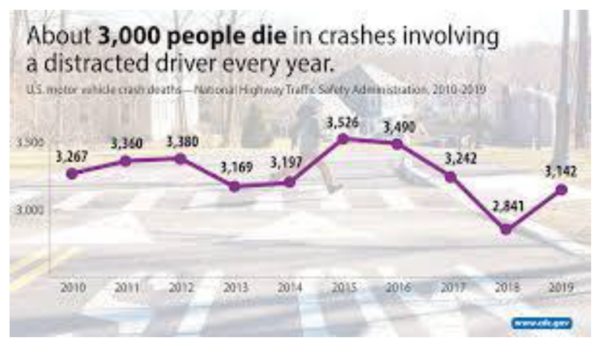
PURPOSE

Distracted driving is responsible for about **3,000** deaths and over **400,000** injuries yearly in the US alone. The purpose of our project is to explore how CNNs and multilayer perceptron models (MLP) can be used to detect distracted driving most effectively.





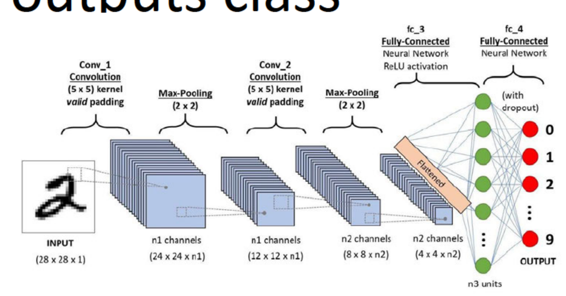
BACKGROUND

Various countries are introducing policies for requiring manufacturers to incorporate **driver-monitoring systems** into their new cars. These systems detect inattentiveness, alert drivers, and allow for automotive intervention when necessary. **CNNs are the most effective models** that can be utilized for these systems. Research suggests that advanced CNN architectures such as MobileNetV2 will allow these systems to provide both extremely **high accuracy and levels of safety** on the road.

We choose a State Farm Distracted Driver dataset. Images in the dataset are taken inside vehicles and display **individuals** doing multiple different actions considered “distractive” (e.g. texting, eating, speaking on the phone). The dataset contains over **4 GB** of data and has over **102152** images of distracted and non distracted drivers portrayed inside their vehicle.

METHODS

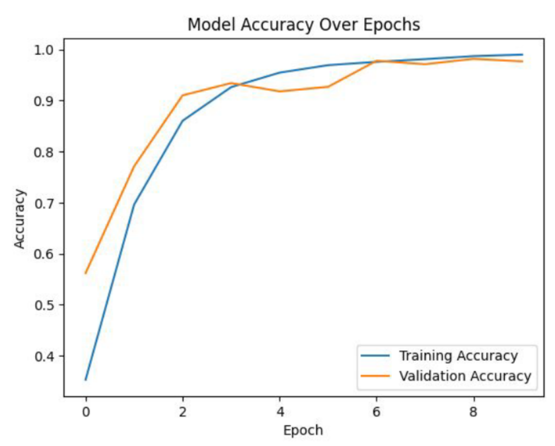
The **CNN** architecture includes a convolutional layer with 8 filters and a 3×3 kernel, followed by 2×2 max pooling and ReLU activation. The output is flattened and passed through a dense layer with 100 neurons before a softmax layer outputs class probabilities



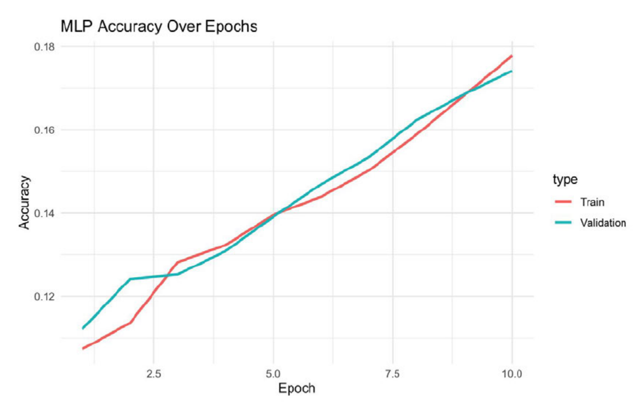
The **MLP** architecture begins with the forward pass with a linear transformation applied followed by ReLU activation and a softmax output. Model performance was evaluated using cross-entropy loss, and gradients were computed via backpropagation

RESULTS

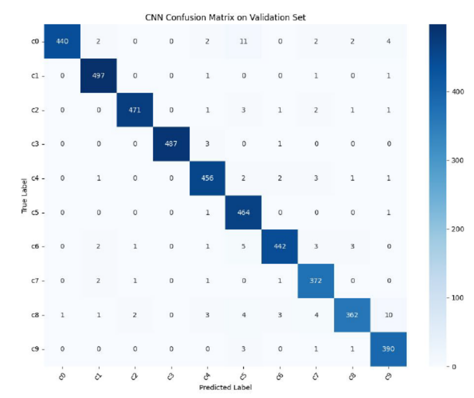
The CNN achieved a final validation accuracy of approximately 98.95%, whereas the MLP plateaued at a considerably lower accuracy of 17.41%. The CNN's performance is not only higher in absolute terms but is also accompanied by consistent convergence patterns, strong per-class performance, and robust generalization.



CNN training and validation accuracy increase sharply in the four epochs and diminishing returns afterwards.



MLP model accuracy increases marginally and consistently over epochs.



CNN confusion matrix shows strong per-class performance.

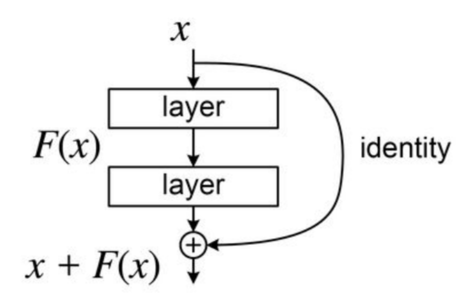
- The CNN achieved precision, recall, and F1-scores all equal to 0.98. Class-level metrics were similarly high: for example, “texting left” (c3) achieved an F1-score of 1.00. The lowest recall occurred in “hair and makeup” (c8), at 0.93, which is consistent with its tendency to be confused with adjacent visual activities.
- The MLP’s accuracy of 17.4% is only slightly better than the No Information Rate (NIR) of 10.95% and Cohen's Kappa score was 0.0748, indicating poor agreement between predicted and true labels.

CONCLUSION

The **CNN model significantly outperformed** the MLP model (final validation accuracy of 98.95% vs. 17.41%). The CNN is effective for driver distraction detection tasks when combined with high-quality datasets. This project gave us substantial practice working with image data and **creating models with practical and meaningful applications in mind**.

NEXT STEPS

- Evaluate how well the model performs across different drivers, for example explore how gender, age affects results and to reduce bias
- Explore more advanced CNN architectures such as MobileNetV2, ResNet, or EfficientNet to see if results differ
- Train on larger images (ex. 128x128) to improve the model’s ability to capture fine-grained features like facial expressions or finger movements



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